

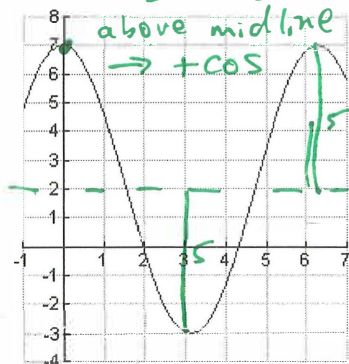
AP Precalculus
Unit 2 Wkst A

Name: _____

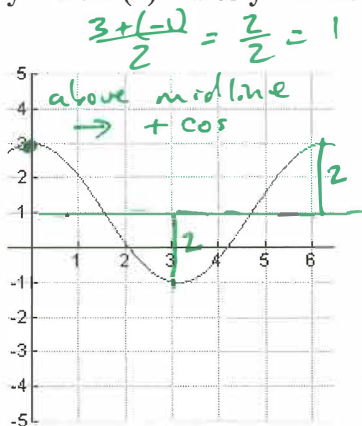
A) For each of the following graphs, write the equation of the midline and find the amplitude.

B) Then write an equation in the form $y = a \sin(x) + d$ or $y = a \cos(x) + d$

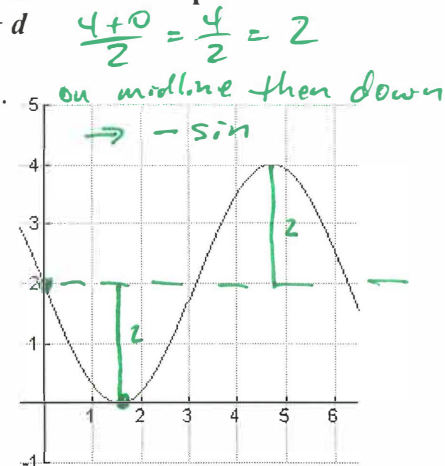
1. $\frac{7+(-3)}{2} = \frac{4}{2} = 2$



2.



3.



Midline: $y = 2$ $d = 2$

Midline: $y = 1$ $d = 1$

Midline: $y = 2$ $d = 2$

Amplitude: 5 $a = +5$

Amplitude: 2 $a = 2$

Amplitude: 2.5 $a = -2.5$

Equation: $y = 5 \cos(x) + 2$

Equation: $y = 2 \cos(x) + 1$

Equation: $y = -2.5 \sin(x) + 2$

For each of the following functions find the (a) midline, (b) amplitude, (c) length of the period, and (d) boundaries of one period. Then (e) graph one period of the function on graph paper.

4. $y = 4 \cos\left(x + \frac{\pi}{4}\right) - 1$

6. $y = 3 \sin\left(2x - \frac{\pi}{2}\right) + 2$

5. $y = -\sin\left(\frac{2\pi}{3}x\right) + 5$

7. $y = 3 \cos(2x - \pi) - 2$

you will get feedback on a-d, plus 5 x-values from Delta Math Practice 1.
see below for the shapes of the graphs

Describe the transformation (translation/shift, dilation/stretch-shrink, reflection) of $g(x)$ compared to $f(x)$. Be specific. Make sure to tell direction and amount of the transformation.

8. $f(x) = \sin x$
 $g(x) = 5 \sin(x - \pi)$

10. $f(x) = \sin x$
 $g(x) = \sin 4x$

9. $f(x) = \cos x$
 $g(x) = -\cos x - 3$

11. $f(x) = \cos x$
 $g(x) = 8 - \cos x$

see below

8. vertical dilation by a factor of 5 (stretch)
horizontal translation (shift) π units to the right

9. reflection across the midline
vertical translation (shift) 3 units down

10. horizontal dilation by a factor of $\frac{1}{4}$ (shrink)

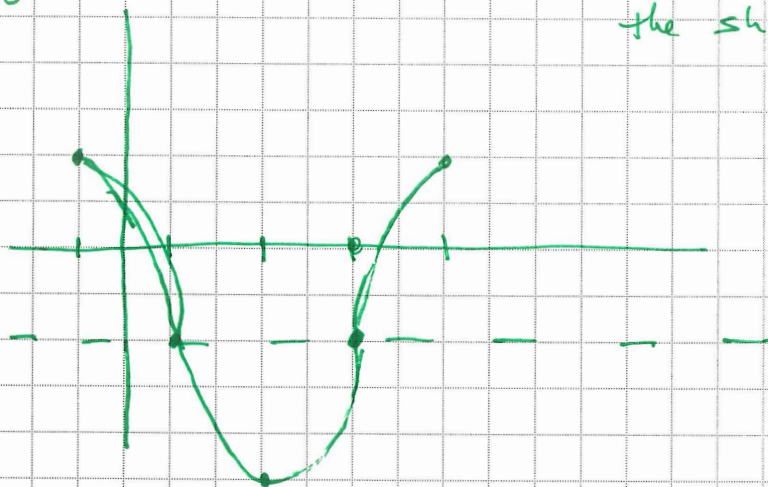
11. rewrite as: $-\cos x + 8$

reflection across the midline

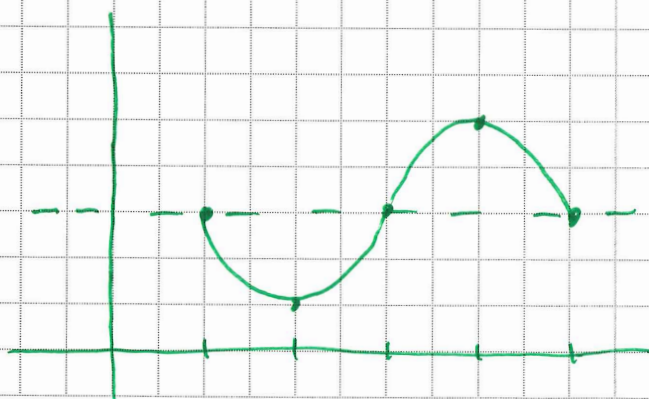
vertical translation (shift) 8 units up

graphs not drawn to scale. use to check
the shape of your graph

4.

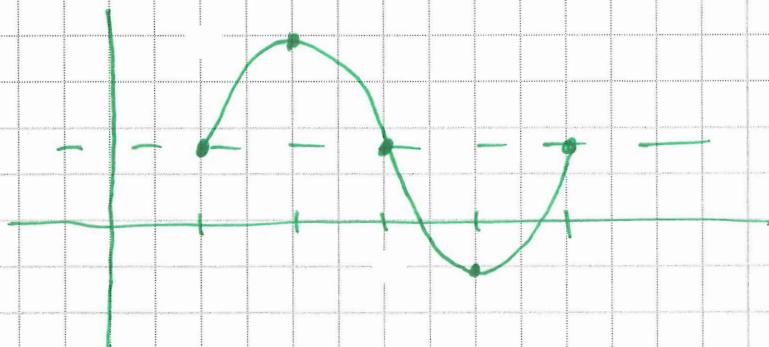


5.



graphs not drawn to scale
use these to check the shape of your graph

6.



7.

